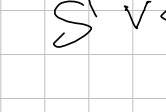


$ab=ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$

 $S^1 \vee S^1$


 $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$
 $T: \mathbb{Z} \times \mathbb{Z}$
 $T \circ \mathbb{Z} \times \mathbb{Z}$
 $f: \mathbb{Z} \times \mathbb{Z}$
 4. proven

[illegible]


 $S: \begin{array}{l} 0\text{-cell } 1 \\ 1\text{-cell } 1 \\ 2\text{-cell } 1 \end{array}$


 $S: \begin{array}{l} 0\text{-cell } 1 \\ 1\text{-cell } 1 \end{array}$


 $S: \begin{array}{l} 0\text{-cell } 1 \\ 1\text{-cell } 1 \end{array}$


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 $S: \begin{array}{l} 0\text{-cell } 1 \\ 1\text{-cell } 1 \end{array}$

Handwritten notes and diagrams on a grid background:

- Top left: A circle with a horizontal line segment inside, and a vertical line segment connecting the top and bottom of the circle. A small dot is at the top of the vertical segment.
- Top center: A large arrow pointing right, followed by the text $\mathbb{R}^n \times \mathbb{R}^n$ and $\mathbb{R}^n$.
- Top right: A large arrow pointing right, followed by the text $\mathbb{R}^n \times \mathbb{R}^n$ and $\mathbb{R}^n$.
- Bottom left: A five-petaled flower-like shape.
- Bottom center: A large arrow pointing right, followed by the text $\mathbb{R}^n \times \mathbb{R}^n$ and $\mathbb{R}^n$.
- Bottom right: A large arrow pointing right, followed by the text $\mathbb{R}^n \times \mathbb{R}^n$ and $\mathbb{R}^n$.
- Far right: A diagram of a square with a circle inside, and a small dot at the top of the circle.

$z = x + iy$
 $\downarrow \text{ex}$
 $K_1 = \frac{1}{4} \cdot 2$
 $\frac{1}{4} \cdot 2$